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SUPERSEDING
Ordnance Department
Specification AXS-1818
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MILITARY SPECIFICATION

DRAWINGS AND DATA LISTS, REPRODUCIBLE; PREPARATION OF

(For Tank and Automotive Materiel)

1. SCOPE

1.1 Scope. - This specification covers requirements for the preparation of original drawings, tracings, data lists, and reproducible copies thereof, for tank and automotive materiel.

1.2 Types and classes. - Drawings and data lists shall be of the following types and classes as specified (see 6.2):

Type I. - Original drawings or tracings.

Class A. Lead pencil or ink on tracing cloth.
Class B. Lead pencil or ink on bond tracing paper.
Class C. Lead pencil or ink on vellum tracing paper.

Type II. - Duplicate drawings (processed tracings).

Class A. Photographic reproduction on cloth.
Class B. Photographic reproduction on paper.

Type III. - Brownline positives (vandykes).

Class A. Cloth brownline prints, direct reading.
Class B. Cloth brownline prints, indirect reading.
Class C. Paper brownline prints, direct reading.
Class D. Paper brownline prints, indirect reading.

Type IV. - Vandyke negatives.

Class A. Cloth vandyke negatives, direct reading.
Class B. Cloth vandyke negatives, indirect reading.
Class C. Paper vandyke negatives, direct reading.
Class D. Paper vandyke negatives, indirect reading.

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Type V. - Other photographic processes, positives.

Type VI. - Data lists.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications. - The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATIONS

DDD-C-471	-	Cloth; Sensitized.
TT-I-528	-	Ink; Drawing, Waterproof, Black.
UU-P-79	-	Paper; Blueprint (Sensitized and Unsensitized).
UU-P-141	-	Paper; Brownprint (Sensitized, and Sensitized and Transparentized).

MILITARY SPECIFICATION

MIL-V-3329	-	Vehicles, Combat, Self-Propelled and Towed; General Requirements for.

U. S. ARMY SPECIFICATION

100-2	-	Standard Specification for Marking Shipments by Contractors.

2.2 Other publication. - The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

ORDNANCE CORPS PUBLICATION

Drafting Room Regulations, Ordnance Corps, U. S. Army.

(Copies of specifications and other publications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Materials. - The materials (paper, ink, pencils, etc.) used in the production of drawings and lists shall be as specified herein. When not specified, such materials shall be of high-grade commercial quality approved by the procuring agency.

3.2 Drawings. - The drawings shall be orthographic projection (third angle) drawings, except that when specified (see 6.2) perspective drawings shall be produced.

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3.3 Data lists. - Data lists as specified herein shall include alphabetical or numerical indexes, engineering parts lists, combined lists of all parts, spare parts, equipment, and tools, or other data lists as specified (see 6.2).

3.4 Type I drawings. - Original drawings or tracings shall be prepared in accordance with instructions contained in Ordnance Corps Drafting Room Regulations (see 6.3) and/or instructions furnished by the procuring agency. They shall be clearly drawn or traced with distinct lines and well-formed characters of adequate opacity to reproduce type II, III (in necessary successive steps), IV, and V drawings meeting the legibility requirements (see 3.10).

3.4.1 Class A. - Drawings shall be drawn and/or traced on tracing cloth (ink and pencil) No. 166 Keuffel and Esser ("Phoenix") or approved equal.

3.4.2 Class B. - Drawings shall be drawn and/or traced on bond tracing paper (ink and pencil) No. 167 Dietzgen ("Briston") or No. 174X Keuffel and Esser ("Banknote") or approved equal.

3.4.3 Class C. - Drawings shall be drawn and/or traced on vellum tracing paper (ink and pencil) No. 195TX Keuffel and Esser ("Albanine") or approved equal.

3.5 Type II drawings. - Duplicate drawings (processed tracings) shall be photographic reproductions capable of reproducing type IV drawings (vandyke negatives) or type V drawings (black line or red line positives) providing the legibility required in 3.10. When tested as specified in 4.2.1, duplicate drawings shall reproduce vandyke negatives on a continuous electric blueprint machine without sorting for machine or printing speed and without requiring individual adjustments of machine or printing speed to secure satisfactory reproductions. They may be of the darkroom or non-darkroom type, unless otherwise specified, and shall be washed, fixed, and processed as necessary to insure keeping qualities consistent with the maximum for the material, without such discoloration, brittleness, or loss of transparency as would affect legibility and reproducibility. They may be of the contact or projection type; but if of the projection type, the duplicate drawings shall be full-size reproductions of the original drawings.

3.5.1 Class A. - Duplicate drawings shall be reproduced on sensitized cloth conforming to type I, class A or B of Specification DDD-C-471.

3.5.2 Class B. - Duplicate drawings shall be reproduced on high-grade commercial sensitized tracing paper.

3.6 Type III drawings. - Brownline positives (vandykes) shall show dark brown lines on a white background free from such reproducible discoloration as would affect legibility. They shall meet the legibility requirements

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of 3.10. When tested as specified in 4.2.2, brownline positives shall reproduce acceptable blueprints on blueprint paper conforming to type I of Specification UU-P-79 and type IV drawings (vandyke negatives) meeting the requirements of 3.7 on a continuous electric blueprint machine without sorting for machine or printing speed and without requiring individual adjustments of machine or printing speed to secure satisfactory reproduction.

3.6.1 Class A. - Brownline positives shall be reproduced on sensitized cloth conforming to type III of Specification DDD-C-471 and shall be direct reading.

3.6.2 Class B. - Brownline positives shall be reproduced on sensitized cloth conforming to type III of Specification DDD-C-471 and shall be indirect reading.

3.6.3 Class C. - Brownline positives shall be reproduced on 24- or 28-pound brownprint paper conforming to type II of Specification UU-P-141 and shall be direct reading.

3.6.4 Class D. - Brownprint positives shall be reproduced on 24- or 28-pound brownprint paper conforming to type II of Specification UU-P-141 and shall be indirect reading.

3.7 Type IV drawings. - Vandyke negatives shall be white lines on a dark brown background free from such reproducible discoloration as would affect legibility. They shall meet the legibility requirements of 3.10. When tested as specified in 4.2.3, vandyke negatives shall reproduce type III drawings (brownline positives) meeting the requirements of 3.6 on continuous electric blueprint machine without sorting for machine or printing speed and without requiring individual adjustments of machine or printing speed to secure satisfactory reproduction.

3.7.1 Class A. - Vandyke negatives shall be reproduced on sensitized cloth conforming to type III of Specification DDD-C-471 and shall be direct reading.

3.7.2 Class B. - Vandyke negatives shall be reproduced on sensitized cloth conforming to type III of Specification DDD-C-471 and shall be indirect reading.

3.7.3 Class C. - Vandyke negatives shall be reproduced on 24- or 28-pound brownprint paper conforming to type II of Specification UU-P-141 and shall be direct reading.

3.7.4 Class D. - Vandyke negatives shall be reproduced on 24- or 28-pound brownprint paper conforming to type II of Specification UU-P-141 and shall be indirect reading.

3.8 Type V drawings. - Positives made by other photographic processes shall be either black or red (maroon) lines on a white background, and shall

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be direct reading. They shall be of high-grade commercial quality and shall be free from such reproducible discoloration as would affect legibility. They shall meet the legibility requirements of 3.10. When tested as specified in 4.2.4, black line or red line positives shall reproduce acceptable blueprints on blueprint paper conforming to type I of Specification UU-P-79 and type IV drawings (vandyke negatives) meeting the requirements of 3.7 on a continuous electric blueprint machine without sorting for machine or printing speed and without requiring individual adjustments of machine or printing speed to secure satisfactory reproduction.

3.9 Type VI data lists. - Data lists shall be original typewritten copy, on high-grade bond paper, of such legibility that it will withstand a two-step vandyke process of reproduction, while maintaining the legibility called for in 3.10.

3.10 Legibility. - All types of drawings and data lists shall be of such legibility that a minimum of 50 readable reproductions can be made therefrom.

3.11 Workmanship. - Drawings and data lists shall be accurate and free from imperfections which may affect readability and reproducibility. Workmanship shall be first-class.

4. INSPECTION PROCEDURES

4.1 Inspection. - Drawings and data lists under this specification shall be subject to inspection in accordance with the applicable provisions of Specification MIL-V-3329.

4.2 Proof prints (checking prints). - When proof prints are required for inspection (see 6.2) they shall be furnished in accordance with the following:

4.2.1 Types I and II. - Proof prints shall be blueprints reproduced from direct reading brownline positives (type III, class C). These brownline positives shall be reproduced from vandyke negatives which shall be reproduced from type I or type II drawings.

4.2.2 Type III and V. - Proof prints shall be blueprints reproduced from type III brownline positives or type V black line or red line positives.

4.2.3 Type IV. - Proof prints shall be blueprints reproduced from direct reading brownline positives (type III, class A) which shall be reproduced from type IV vandyke negatives.

4.2.4 Type VI. - Proof data lists shall be blueprints reproduced by any two-step vandyke process specified herein.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - Drawings and data lists shall be arranged numerically by size and shall not be folded. Large shipments shall be packaged flat or rolled. Drawings rolled for shipment shall be rolled on a rigid core which will extend beyond the ends of the rolls sufficiently to eliminate possible injury to drawings. In no case shall any drawing on the roll be larger than any other drawing between it and the rigid core. Small shipments of types III, IV, or V only may be rolled without the use of a rigid core. Each bundle (if packaged flat) and roll shall be wrapped securely in waterproof and lightproof material.

5.2 Packing.

5.2.1 For domestic shipment. - Unless otherwise specified, interior wraps or packages shall be packed in substantial commercial exterior shipping containers so constructed as to insure acceptance by common or other carriers for safe transportation, at the lowest rate, to the point of delivery specified in the contract or purchase order.

5.2.3 For oversea shipment. - Packing for oversea shipment (see 6.2) shall be in accordance with instructions furnished by the procuring agency in connection with the contract or purchase order.

5.3 Marking. - Unless otherwise specified, marking for shipment shall be in accordance with Specification 100-2.

6. NOTES

6.1 Intended use. - This specification is intended for use in procuring drawings, tracings and related data, including the preparation thereof, as required for the design, manufacture and maintenance of tank and automotive materiel.

6.2 Ordering data. - Contracts or orders should specify the following information:

a. Title, number and date of this specification.
b. Type and class of drawings. (See 1.2).
c. When perspective drawings are required. (See 3.2).
d. Data lists to be furnished. (See 3.3).
e. List of proof (checking) prints, when required. (See 4.2).
f. Instructions for oversea shipment. (See 5.2.3).

6.3 Regulations. - In order to permit the contractor to comply with the requirements of 3.4 of this specification, the procuring agency should arrange to lend the contractor a copy of Ordnance Corps Drafting Room Regulations for his use during the period of the contract.

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NOTICE. - This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any purpose except when specifically authorized by the Chief of Ordnance.

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